Amplitude-Weighted Cyclicity and Laurent Moment Channels

Persistent moment vanishing for every polynomial phase of fixed-seed rank at most three

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Abstract

Let $P \in \mathbf{C}[x]$ be a polynomial phase, let $A(x)\,dx$ be a polynomial amplitude, and let $G' = A$. We prove that the cyclic rank generated by $[A\,dx]$ in the exponential twisted de Rham module is exactly the number of non-trace residue channels in $G(\xi(t))$, where $P(\xi(t)) = t^{\deg P}$. Equivalently, it is the reduced constant span of the primitive values $G(x_j)$ on the generic inverse roots. This extends the fixed-seed Atlas from dx to every polynomial amplitude and gives the sharp rank-zero criterion $G \in \mathbf{C}[P]$. A discrete Fourier transform then classifies vanishing on every specified zero-dimensional cycle.

For endpoint moments

$$\int_a^b P(x)^k A(x)\,dx = 0 \qquad (k \geq 0),$$

we combine the Pakovich–Muzychuk solution of the polynomial moment problem with the all-degree right-factor audit in the preceding Atlas. Conditional only on that still-unrefereed Atlas classification, we obtain an explicit family-by-family answer for every phase with fixed-seed rank $q(P) \leq 3$: powers, Dickson polynomials, quadratic- and cubic-over-power phases, indecomposable quartics, and the exceptional degree-twelve collision.

For a Laurent phase L with pole order m at zero and an arbitrary Laurent amplitude $A(z)\,dz$, a local inverse $L(\phi(\tau)) = \tau^{-m}$ gives the exact criterion

$$\operatorname{Res}_0 L^k A\,dz = [\tau^{mk-1}]A(\phi(\tau))\phi'(\tau).$$

This includes logarithmic amplitudes and reduces exact forms to a class-zero coefficient test. It does not assert a finite normal-form classification for all Laurent phases: known indecomposable examples have persistent solutions beyond a composition-only ansatz. Exact ordinary and optimised replays, semantic mutations, and direct residue checks accompany the proof. The result is an unrefereed theorem candidate, not external peer review, a complete formalisation, or a priority claim.

中文摘要

令 $P \in \mathbf{C}[x]$ 為多項式相位，$A(x)\,dx$ 為多項式振幅，且 $G' = A$。本文證明：扭曲 de Rham 模中由 $[A\,dx]$ 生成的循環秩，恰等於 $G(\xi(t))$ 的非跡同餘通道數，其中 $P(\xi(t)) = t^{\deg P}$；亦即，它 等於一般反根上 $G(x_j)$ 的約化常數張成維數。因此固定種子 dx 的圖譜 可推廣到所有多項式振幅，且秩為零當且僅當 $G \in \mathbf{C}[P]$。離散 Fourier 變換進一步把指定零維循環上的恆等消失 化為精確的支撐消去條件。

對端點矩 $\int_a^b P^k A\,dx = 0$，本文把 Pakovich–Muzychuk 的多項式矩 問題定理與前一版 Atlas 的全次數右因子審計結合。於該尚未經同儕審查的 Atlas 分類條件下，我們逐族分類所有 $q(P) \leq 3$ 的相位，並保留 Dickson 與特殊碰撞中不可比較右因子之和。對 Laurent 相位 L，任意 Laurent 一形式 $A(z)\,dz$ 的矩由局部反函數係數 $[\tau^{mk-1}]A(\phi)\phi'$ 精確決定。已知反例表明一般 Laurent 解不能僅 以簡單複合解描述，故本文不宣稱有限正規形圖譜。

Candidate status and additive lineage

> **Version 0.7.0-candidate.** This is an additive research successor. It neither deletes nor supersedes the two original Evidence Press papers or the all-degree Atlas. Its endpoint classification is relative to the Pakovich–Muzychuk theorem and to the Atlas's unrefereed all-degree right-factor classification.

The preserved predecessors are:

1. *Fixed-Seed Cyclicity Loci for Polynomial Exponential Periods* [EP1];
2. *Fixed-seed cyclic rank as reduced inverse-root span* [EP2]; and
3. *Inverse-Root Support and Fourier Fusion for Polynomial Exponential Periods: All-Degree Low-Rank Completion* [EP3].

The present paper changes the seed, not the history. It replaces dx by an arbitrary polynomial amplitude, specialises a published moment theorem through the Atlas, and adds a rigorously bounded Laurent-phase result. The source package retains the predecessor PDFs byte-for-byte.

1. Three notions of persistent vanishing

We distinguish three questions.

1. **Amplitude cyclicity.** What is the rank generated by $[A\,dx]$ under the exponential connection for a fixed polynomial phase P?
2. **Endpoint or zero-cycle moments.** For which amplitudes do all moments vanish between fixed endpoints, or on a fixed zero-dimensional cycle?
3. **Laurent residues.** For which Laurent one-forms do all residues $\operatorname{Res}_0 L^k A\,dz$ vanish?

These questions interact, but they are not identical. In particular, $q(P, dx) \le 3$ does not imply that every amplitude has cyclic rank at most three.

Let $d = \deg P \ge 2$, put $K = \mathbf{C}(s)$, and write

$$\mathcal{H}_P = K[x]\,dx/(\partial_x + sP')K[x]\,dx. \tag{1.1}$$

The connection is

$$\nabla_s[f\,dx] = [(\partial_s f + Pf)\,dx]. \tag{1.2}$$

For $A \in \mathbf{C}[x]$, define

$$q(P, A\,dx) = \dim_K K\langle\nabla_s\rangle[A\,dx]. \tag{1.3}$$

An affine source change transports both P and $A\,dx$; an affine target change acts triangularly on powers of P. We may take P monic and centred when computing support.

2. Primitive-amplitude support

Let

$$P(\xi(t)) = t^d, \qquad \xi(t) = t + O(t^{-1}) \tag{2.1}$$

be the normalised inverse at infinity, and let $G' = A$. Write

$$G(\xi(t)) = \sum_{n \le N} c_n t^n. \tag{2.2}$$

The **primitive-amplitude support** is

$$\Sigma(P; G) = \{r \in \{1, \dots, d-1\} : c_n \text{ is nonzero for some } n \equiv r \pmod d\}. \tag{2.3}$$

Changing the additive constant in G changes only the trace channel.

Theorem 2.1 — Polynomial-amplitude support theorem

For every polynomial phase and polynomial amplitude,

$$\boxed{q(P, A\,dx) = |\Sigma(P; G)| = \dim_{\mathbf{C}} \operatorname{span}^{\mathrm{red}}\{G(x_0), \dots, G(x_{d-1})\}.} \tag{2.4}$$

Here $x_j = \xi(\zeta_d^j t)$, and "red" removes the common trace component.

Proof

The formal moment isomorphism of [EP3] is reproduced in the form needed here. Pull a one-form back by $x = \xi(t)$. In nonzero residue class $r \pmod d$, send

$$t^{r+nd-1}dt \longmapsto \mu_{r,n} s^{-n} e_r, \qquad \mu_{r,n+1} = -\frac{r+nd}{d}\mu_{r,n}. \tag{2.5}$$

All $\mu_{r,n}$ are nonzero. Equation $P(\xi) = t^d$ makes $(\partial_x + sP')h\,dx$ pull back to $(\partial_t + sdt^{d-1})h(\xi(t))\,dt$, which maps to zero by (2.5). The map descends to

$$\mathcal{H}_P \otimes_K \mathbf{C}((s)) \longrightarrow \bigoplus_{r=1}^{d-1} \mathbf{C}((s))e_r. \tag{2.6}$$

On $dx, x\,dx, \dots, x^{d-2}dx$, its constant matrix is triangular with diagonal one because $\xi^j \xi' = t^j + O(t^{j-2})$. Hence (2.6) is an isomorphism.

Now

$$A(\xi)\xi' dt = d(G(\xi)) = \sum_n n c_n t^{n-1} dt. \tag{2.7}$$

For a nonzero residue r, n cannot vanish when $n \equiv r \pmod d$. Distinct coefficients occupy distinct powers of s, so the r-th component is nonzero exactly when $r \in \Sigma(P;G)$. Active Kummer components have distinct fractional exponents; their derivative Wronskian is a nonzero Vandermonde determinant [EP3, Lemma 5.1]. Their cyclic rank is their number.

Finally,

$$Y_r(t) = \frac{1}{d}\sum_{j=0}^{d-1} \zeta_d^{-jr} G(x_j(t))$$

is the part of (2.2) in residue class r. Fourier inversion and disjoint Laurent supports prove the reduced-span equality. □

Corollary 2.2 — Universal rank-zero amplitudes

$$q(P, A\,dx) = 0 \quad\Longleftrightarrow\quad \Sigma(P;G) = \varnothing \quad\Longleftrightarrow\quad G \in \mathbf{C}[P]. \tag{2.8}$$

Equivalently, $A(x) = P'(x)H'(P(x))$ for some polynomial H.

Proof

If $G = H \circ P$, then $G(\xi(t)) = H(t^d)$. Conversely, empty non-trace support makes $G(x_j)$ equal on all generic inverse roots. Hence $G \in \mathbf{C}(P)$ inside $\mathbf{C}(x)$. Because both P and G are polynomials and have no finite pole, $\mathbf{C}(P) \cap \mathbf{C}[x] = \mathbf{C}[P]$. □

Example 2.3 — Low phase rank does not bound amplitude rank

For $P = (x^4 + x)^3$, $q(P, dx) = 3$. Taking $G = x^4$, $A = 4x^3$, gives

$$\Sigma(P; x^4) = \{1, 4, 7, 10\}, \qquad q(P, 4x^3 dx) = 4. \tag{2.9}$$

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3. Exact zero-cycle classification

Let $n = (n_0, \dots, n_{d-1}) \in \mathbf{C}^d$, $\sum_j n_j = 0$, and put

$$\widehat{n}(r) = \sum_{j=0}^{d-1} n_j \zeta_d^{jr}. \tag{3.1}$$

Theorem 3.1 — Support annihilation on a zero-cycle

$$\sum_{j=0}^{d-1} n_j G(x_j(t)) \equiv 0 \quad\Longleftrightarrow\quad \widehat{n}(r) = 0 \text{ for every } r \in \Sigma(P;G). \tag{3.2}$$

Proof

Fourier inversion writes $G(x_j) = \sum_r \zeta_d^{jr} Y_r$. The trace term vanishes because $\widehat{n}(0) = 0$. Nonzero Y_r are constant-linearly independent because their exponents occupy disjoint residue classes. $\square$

This is complete for a specified phase, amplitude, and cycle. It does not make the broader balanced-cycle literature composition-only [ABM13].

For $P = (x^4 + x)^3$, the cycle $n_j = \cos(2\pi j/3)$ has Fourier support $\{4, 8\}$. The primitives

$$G_7 = x^7 + \frac{7}{4}x^4, \qquad G_{10} = x^{10} - \frac{5}{2}x^4 \tag{3.3}$$

both have support $\{1, 7, 10\}$, proving their zero-cycle vanishing. Their amplitudes are $7x^6 + 7x^3$ and $10x^9 - 10x^3$.

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4. Endpoint moments and the imported theorem

Fix $a, b \in \mathbf{C}$, put $G' = A$, and define

$$M_k(P, A; a, b) = \int_a^b P(x)^k A(x)\, dx. \tag{4.1}$$

Theorem 4.1 — Polynomial moment theorem [PM09]

All $M_k(P, A; a, b)$ vanish for $k \geq 0$ if and only if

$$G = \sum_{j=1}^{r} H_j \circ W_j, \tag{4.2}$$

where $P = R_j \circ W_j$ and $W_j(a) = W_j(b)$ for every j.

Necessity is imported from [PM09], not reproved. Sufficiency follows by substitution $u = W_j(x)$. Pakovich's refinement [P13] describes the indecomposable right factors explicitly.

If $P(a)$ and $P(b)$ differ, no nonconstant right factor is endpoint-admissible, so the only solution is $A = 0$. Hence assume $P(a) = P(b)$.

5. Persistent endpoint moments for every $q(P) \le 3$

This section specialises Theorem 4.1 using the all-degree normal forms and right-factor audit of [EP3]. Because [EP3] is unrefereed relative to named monodromy classifications, the same qualification attaches here. Formulas are in Atlas coordinates; affine source changes transport $a, b, A\,dx, G$.

Theorem 5.1 — Low-rank persistent-moment classification

Let $q(P, dx) \le 3$, let a, b be distinct, and let $P(a) = P(b)$. All endpoint moments vanish if and only if G belongs to the final-column space below.

Phase	Endpoint-admissible factors	Complete space for G
$P = x^d$	x^h, where h is the order of a/b	$\mathbf{C}[x^h]$
$P = D_d(x, \alpha)$	divisibility-minimal $D_e(x, \alpha)$ with $e \mid d$ and $D_e(a, \alpha) = D_e(b, \alpha)$	$\sum_{e \in E_{\min}} \mathbf{C}[D_e]$
$P = H(x^m)$, $\deg H = 2$	least x^h, $h \mid m$, with $a^h = b^h$, if one exists; otherwise P	$\mathbf{C}[x^h]$, or $\mathbf{C}[P]$
$P = H(x^m)$, $\deg H = 3$, off the aligned-square locus	the same power factor, if it exists; otherwise P	$\mathbf{C}[x^h]$, or $\mathbf{C}[P]$
$P = H(x^{2k}) = (x^{3k} + bx^k)^2 + C$	P, admissible power factors, and $F = x^{3k} + bx^k$ when $F(a) = F(b)$	$\mathbf{C}[P] + \mathbf{1}_{F(a)=F(b)}\mathbf{C}[F] + \mathbf{1}_{a^h=b^h}\mathbf{C}[x^h]$
indecomposable $P = x^4 + \alpha x^2 + \beta x$, with β nonzero	P only	$\mathbf{C}[P]$
$P = (x^4 + \alpha x)^3$, with α nonzero	P, $W_4 = x^4 + \alpha x$, and $W_3 = x^3$, subject to endpoint equality	$\mathbf{C}[P] + \mathbf{1}_{W_4(a)=W_4(b)}\mathbf{C}[W_4] + \mathbf{1}_{a^3=b^3}\mathbf{C}[x^3]$

Here $E_{\min}$ consists of divisibility-minimal elements of

$$E = \{e : e \mid d,\ D_e(a, \alpha) = D_e(b, \alpha)\}. \tag{5.1}$$

In over-power rows, omit the h-term if no divisor of m has equal endpoint values. Lower-rank overlaps use the earliest applicable row.

Proof

Theorem 4.1 says that the solution space is the sum of $\mathbf{C}[W]$ over endpoint-admissible right factors. For x^d, all factors are x^e, $e \mid d$; endpoint equality holds exactly when $h \mid e$, and their algebras lie in $\mathbf{C}[x^h]$.

The Dickson identity

$$D_d(x, \alpha) = D_{d/e}(D_e(x, \alpha), \alpha^e) \tag{5.2}$$

gives its factors. If $e_1 \mid e_2$, then $\mathbf{C}[D_{e_2}] \subseteq \mathbf{C}[D_{e_1}]$; only minimal admissible factors are needed, while incomparable factors must be summed.

For $H(x^m)$, ordinary suffixes are x^e, $e \mid m$, and collapse to the least admissible x^h. If none is admissible, P remains. The sole additional rank-three suffix is

$$H(x^{2k}) - C = (x^{3k} + bx^k)^2. \tag{5.3}$$

It contributes $\mathbf{C}[F]$ exactly when $F(a) = F(b)$.

An indecomposable quartic has no proper nonlinear factor. The exceptional collision has the incomparable factors $x^4 + \alpha x$ and x^3, since

$$(x^4 + \alpha x)^3 = x^3(x^3 + \alpha)^3 = \widetilde{H}(x^3). \tag{5.4}$$

The table follows from Theorem 4.1. □

Corollary 5.2 — Decision procedure

Affine-normalise P, locate its Atlas row, evaluate the listed factors at a, b, and test whether G belongs to the resulting sum of polynomial subalgebras. This is exact decomposition, not moment sampling.

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6. Laurent phases and arbitrary Laurent amplitudes

Let

$$L(z) = \sum_{i=-m}^{n} c_i z^i, \qquad m, n \geq 1, \tag{6.1}$$

where c_{-m} and c_n are nonzero, and let $A(z)\,dz$ be any Laurent-polynomial one-form. Divide L by c_{-m}. There is a unique formal series $\phi(\tau) = \tau(1 + O(\tau))$ satisfying

$$L(\phi(\tau)) = \tau^{-m}. \tag{6.2}$$

Theorem 6.1 — Laurent local-channel criterion

For every $k \geq 1$,

$$\boxed{\operatorname{Res}_{z=0} L(z)^k A(z)\,dz = [\tau^{mk-1}] A(\phi(\tau))\phi'(\tau).} \tag{6.3}$$

Thus all Laurent moments vanish if and only if all displayed coefficients vanish.

Proof

Formal residues are invariant under $z = \phi(\tau)$. Therefore

$$\operatorname{Res}_{z=0} L^k A\,dz = \operatorname{Res}_{\tau=0} \tau^{-mk} A(\phi(\tau))\phi'(\tau)d\tau,$$

whose τ^{-1} coefficient is (6.3). □

This includes the logarithmic class $z^{-1}dz$.

Corollary 6.2 — Exact Laurent amplitudes

If $A = G'$ for Laurent polynomial G, then

$$\operatorname{Res}_0 L^k \, dG = mk[\tau^{mk}]G(\phi(\tau)). \tag{6.4}$$

If $G = H \circ L$, this vanishes because $G(\phi) = H(\tau^{-m})$. A second mechanism is deck symmetry: if $L = R(z^e)$, excluded residue classes vanish under $z \mapsto \zeta_e z$. For example, $\operatorname{Res}_0(z^2 + z^{-2})^k dz = 0$ for every k. This corrects the inherited exploratory notebook's undercount for symmetric phases.

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7. Why the Laurent result stops here

Theorem 6.1 is exact for specified L, A, but it is not a finite classification of all pairs. Pakovich, Pech, and Zvonkin [PPZ10] exhibit an indecomposable Laurent phase whose persistent solution space is a free module of rank five over $\mathbf{C}[L]$. A simple composition-only description is therefore insufficient.

This release does not infer all-k vanishing from a finite zero prefix, promote inherited small-bidegree tables to a theorem, or claim a finite all-Laurent analogue of Theorem 5.1. It supplies the exact coefficient object against which future proposals must be proved.

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8. Exact computational checks

The implementation compares Krylov reduction of $A, PA, P^2A, \ldots$ with coefficient extraction from $G(\xi(t))$. It stops only when the witnessed support size equals the independently computed cyclic rank. Controls include powers, Dickson phases, affine transport, the rank-four amplitude, and two exceptional zero-cycle centres.

For Laurent inputs, local inverse coefficients are compared with direct Laurent multiplication and residues at every tested moment. Controls include a logarithmic amplitude, deck symmetry, nonvanishing, target scaling, and exact forms.

Ordinary and optimised executions produce byte-identical receipts. Two semantic mutations are rejected after digest recomputation. Sixteen endpoint moments and finite Laurent prefixes remain regressions, not infinite proofs.

9. Assurance boundary and open problems

This manuscript proves the polynomial-amplitude support formula, rank-zero criterion, specified zero-cycle criterion, and arbitrary-Laurent-form local criterion. The endpoint table additionally imports [PM09] and the unrefereed Atlas factor classification [EP3]. It is a complete specialisation in that framework, not an independent reproof of monodromy or Ritt classification.

Next priorities are external specialist review; formalisation of the moment isomorphism; finite structural conditions equivalent to the Laurent test for controlled phases; and classification of amplitudes of bounded induced rank.

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Data and code availability

The source, exact programs, tests, receipt, claims, literature audit, and assurance statement are included in this candidate worktree.

Ethics declaration

This mathematical research uses no human participants, animals, personal data, or sensitive datasets. No ethics approval was required.

Author contributions (CRediT)

Anonymous: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Writing—original draft, and Writing—review and editing. Ian Pitchford served as repository maintainer and Evidence Press publisher; this operational role is not scholarly authorship.

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